

## Foundation (Jalapeno)

### Open Ended Questions (FRQ)

**Q9.** A rocket is traveling at a speed of  $2x + 5$  miles per second. If it needs to travel faster than 25 miles per second, what is the minimum value of  $x$ ? Write your solution in interval notation and graph it on a number line. Required work and graph.

**Q10.** Solve the inequality

$2x + 3 > 7$  and write the solution in interval notation. Then, give a real-world example of how this inequality could be used in a space exploration.

### Chef's Tips

- Provide clear instructions for each question
- Emphasize the importance of showing work and graphing solutions
- Encourage students to use interval notation and number lines to visualize solutions